CMPS453 Cloud Computing Introduction

Ayman AboElHassan, PhD
Assistant Professor
ayman.abo.elmaaty@eng.cu.edu.eg

Background

Brief History of the Internet

1980s: Local Area Networks

- TCP/IP “81”, Email “86”, DNS “84” (Domain Name System)

1990s: WWW public & dotcom bubble

- Yahoo “95”, yellowpages “96”, eBay “95”, google “98”

2000s: Social Networking

- Facebook “2004”, Twitter “2006”, YouTube “2005”

2010s: Cloud Computing

- AWS “2006”, Azure “2010”, GCP “2010”

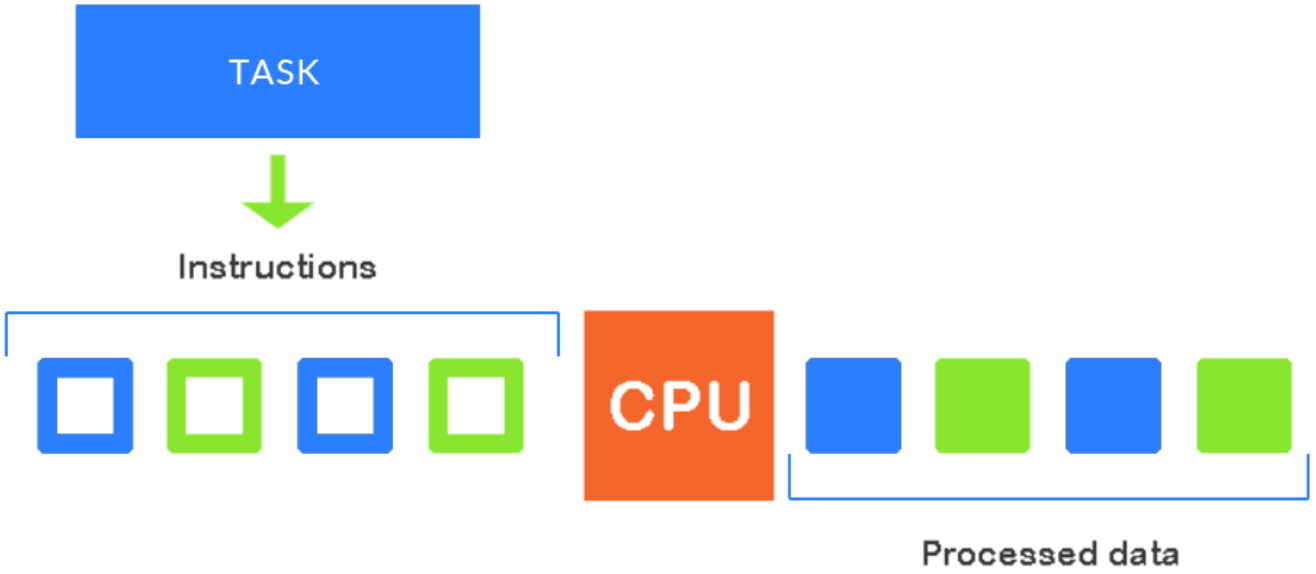
Lot of papers published by these companies about their internal design

2020s: ML & AI Boom

Serial Computing

Old “traditional” software

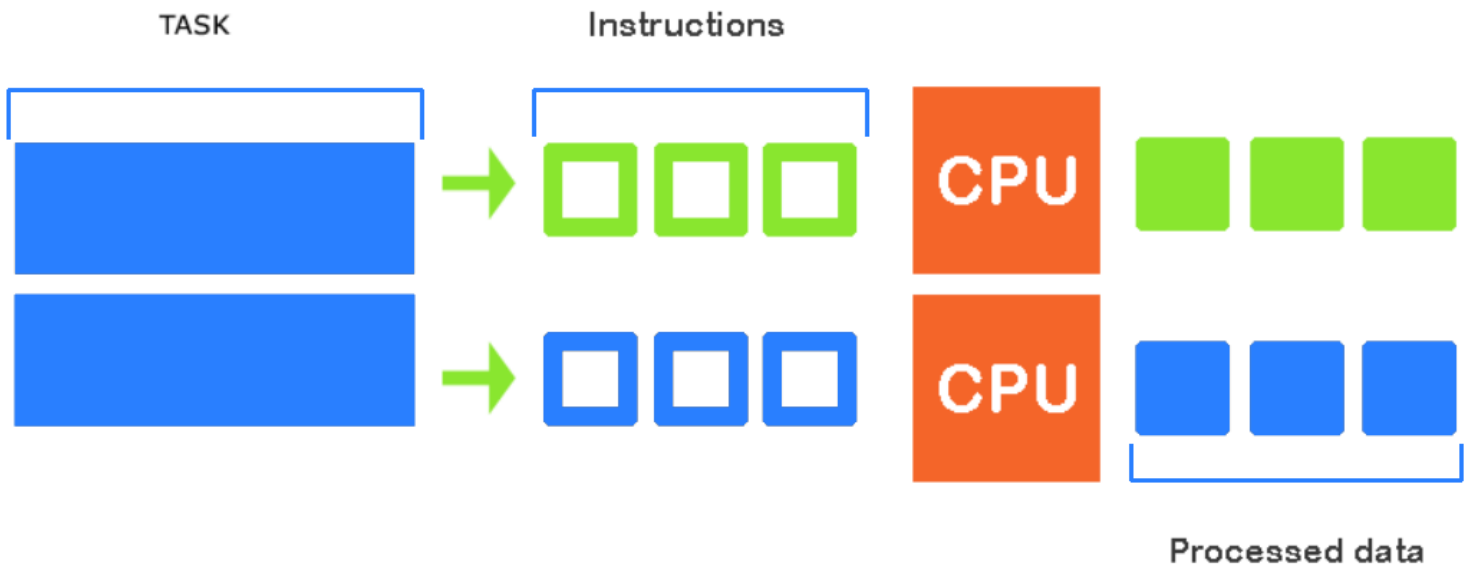
- **Task** is defined as a set of <u>Serial Instructions</u>
- Instructions executed on single-core **CPU** one at a time



Parallel Computing

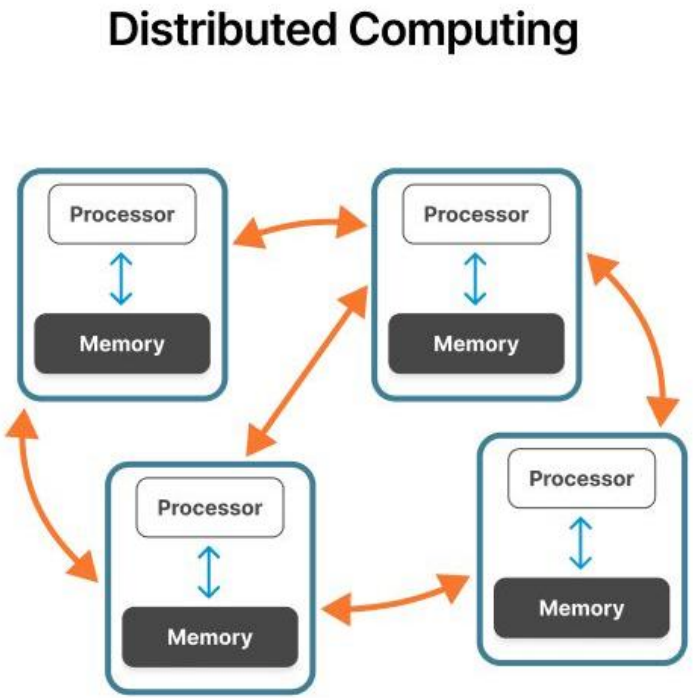
Improve execution performance through parallelism

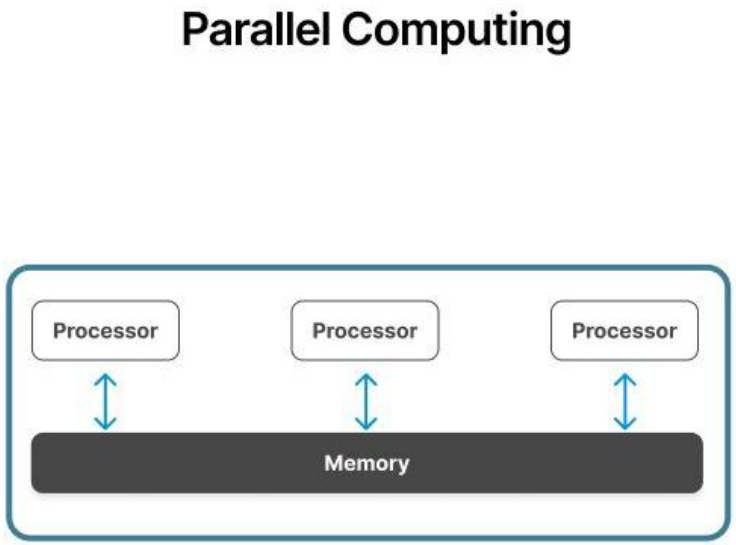
- **Task** is defined as a set of <u>multiple subtasks</u>
- Subtasks are executed on **multiple CPUs** <u>in-parallel</u>



Distributed Computing

Multiple computers connected through a **network cooperating** to perform a service/function concurrently





Why Distributed Computing?

- Do you share code with coworkers using USBs?
- Do you have to physically move to company server to run code?
- Do you need to connect 100 disks to a server?
- What happens if the company server restarts for a windows update & crashes?
- Can you run GTA on company server?

Cloud Computing Concept

Cloud Computing (NIST Definition)

A **model** for enabling ubiquitous, convenient, on-demand network **access** to a *shared pool* of configurable *computing resources* (e.g., networks, servers, storage, applications, and services) that can be **rapidly** provisioned and released with **minimal management effort** or service provider interaction.

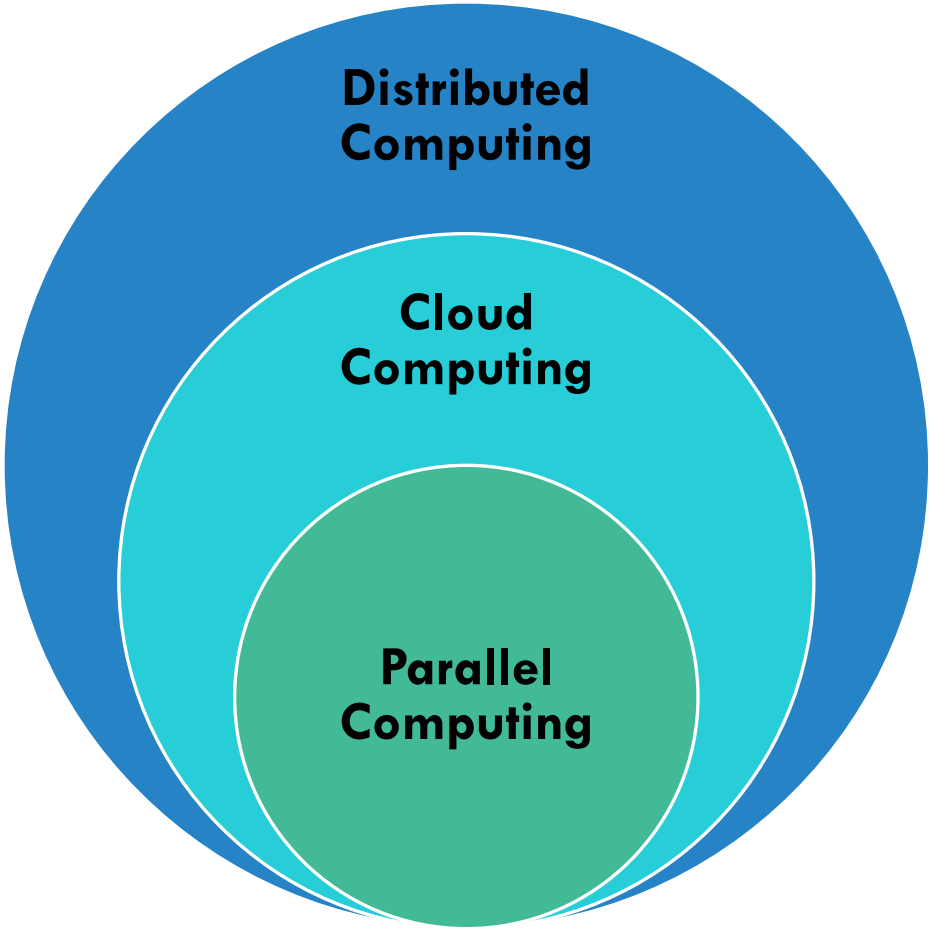
Cloud Computing (NIST Definition)

- Not a technology ➔ It's a **service** delivery model
- Convenient **on-demand** access to IT services
- Requires network **connectivity**
- Rapid resource provisioning ➔ **Elastic** resource management
- Minimal management effort ➔ **User-friendly**
- Minimal service provider interaction ➔ **Self-service**

Cloud Computing

1. Centralized infrastructure
   - Client-Server architecture
2. On-demand resource
3. Highly scalable
4. Accessible through internet

Summary


Distributed Computing
Cloud Computing
Parallel Computing

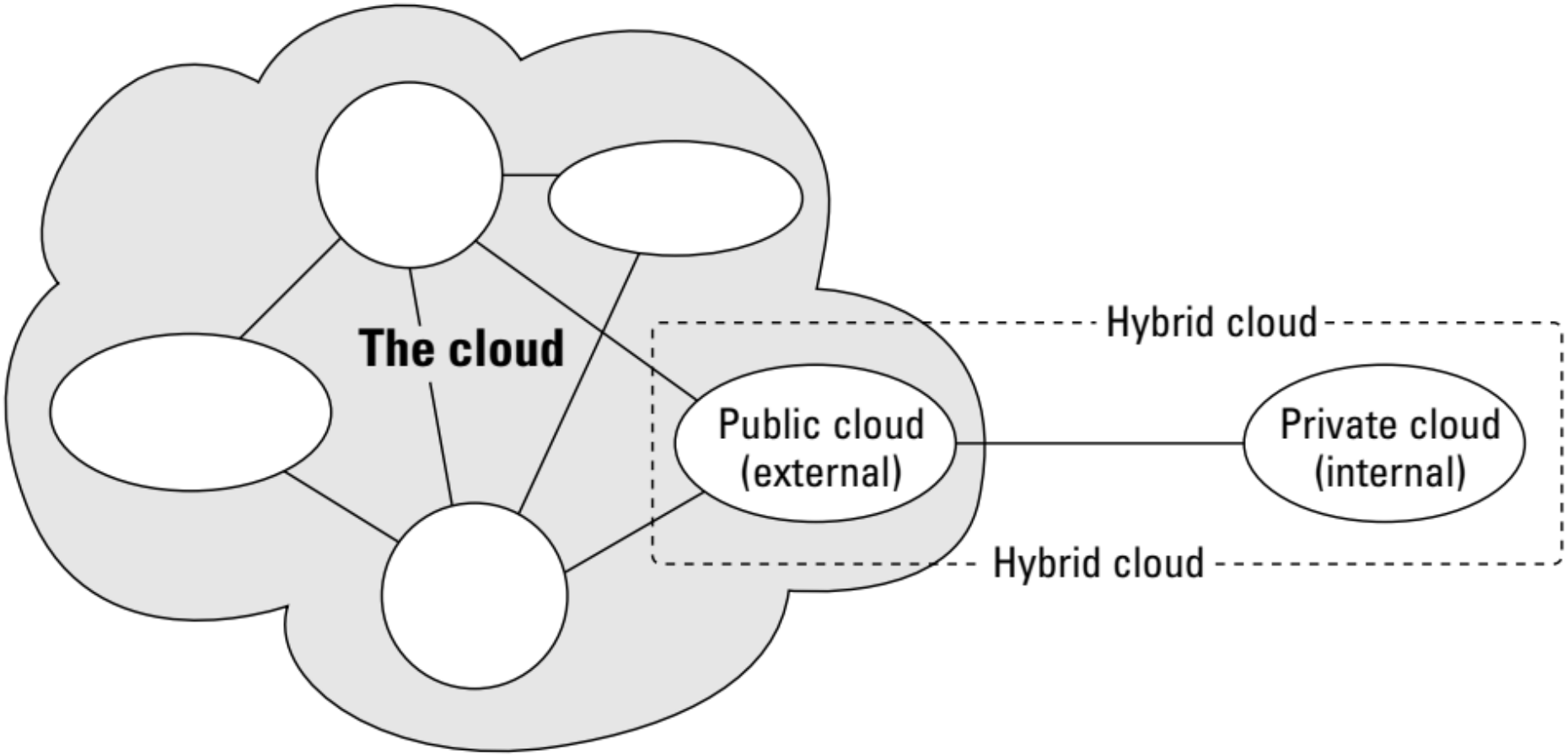
Cloud Computing Models

Cloud Computing Models

2 Perspectives

1. Deployment Models ➔ Where your code deployed?
2. Service Models ➔ What services your code use?

Cloud Deployment Model



Cloud Deployment Model

1. Public Cloud
   - Best-known form of cloud computing
   - Service provider (AWS, Google, …) offer IT services <u>over the internet</u>
   - Clients access cloud resources/services as a pay-per-use subscription

Cloud Deployment Model

1. Public Cloud
2. Private Cloud
   - Deployed for internal use <u>at the enterprise</u> own datacenters
   - Datacenter ingress firewall control access to the data/service
   - Provide better privacy/security

Cloud Deployment Model

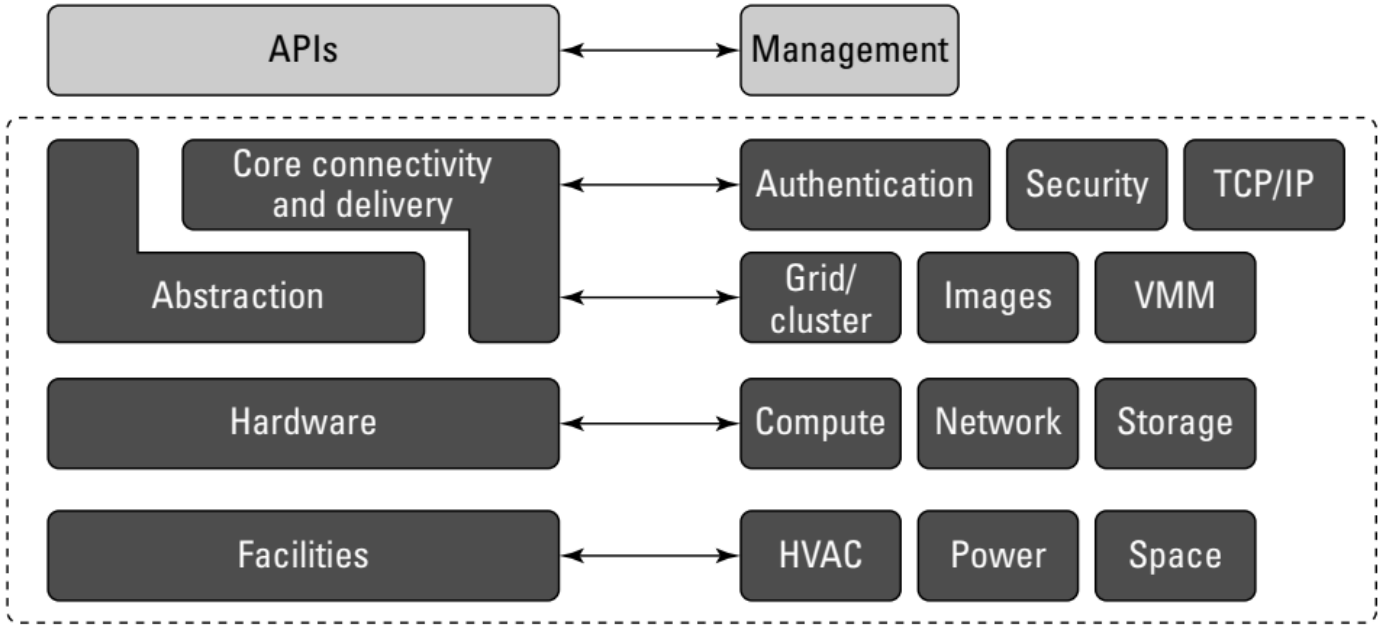
1. Public Cloud
2. Private Cloud
3. Community Cloud
   - Cloud infrastructure is built and managed by specific community
   - Typically have common goal/concern
   - The Bibliotheca Alexandrina Supercomputing facility

Cloud Deployment Model

1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud
   - Flexible cloud deployment mode
   - Comprise two or more different models

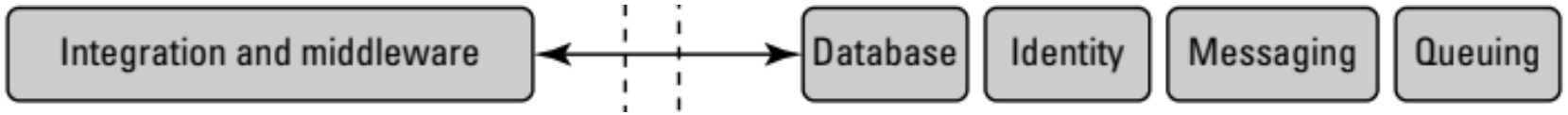
Cloud Service Model

1. Infrastructure as a Service (IaaS)
   - Datacenters provide <u>computational/storage</u> **resources** for clients
   - Clients use these resources as a fully outsourced device



Cloud Service Model

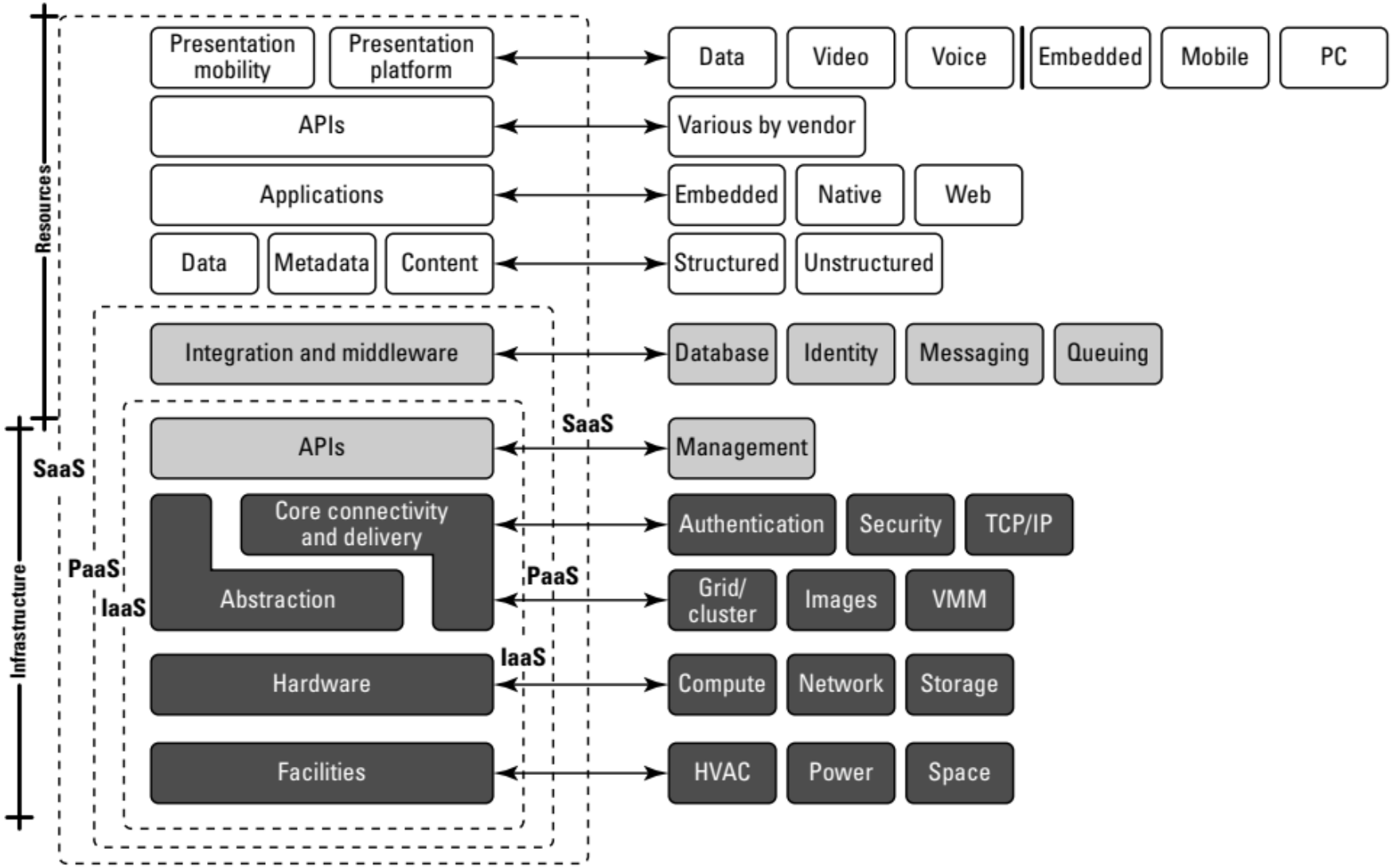
1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
   - A computing platform to support the software life cycle
   - OS/Middleware for **runtime environment**



Cloud Service Model

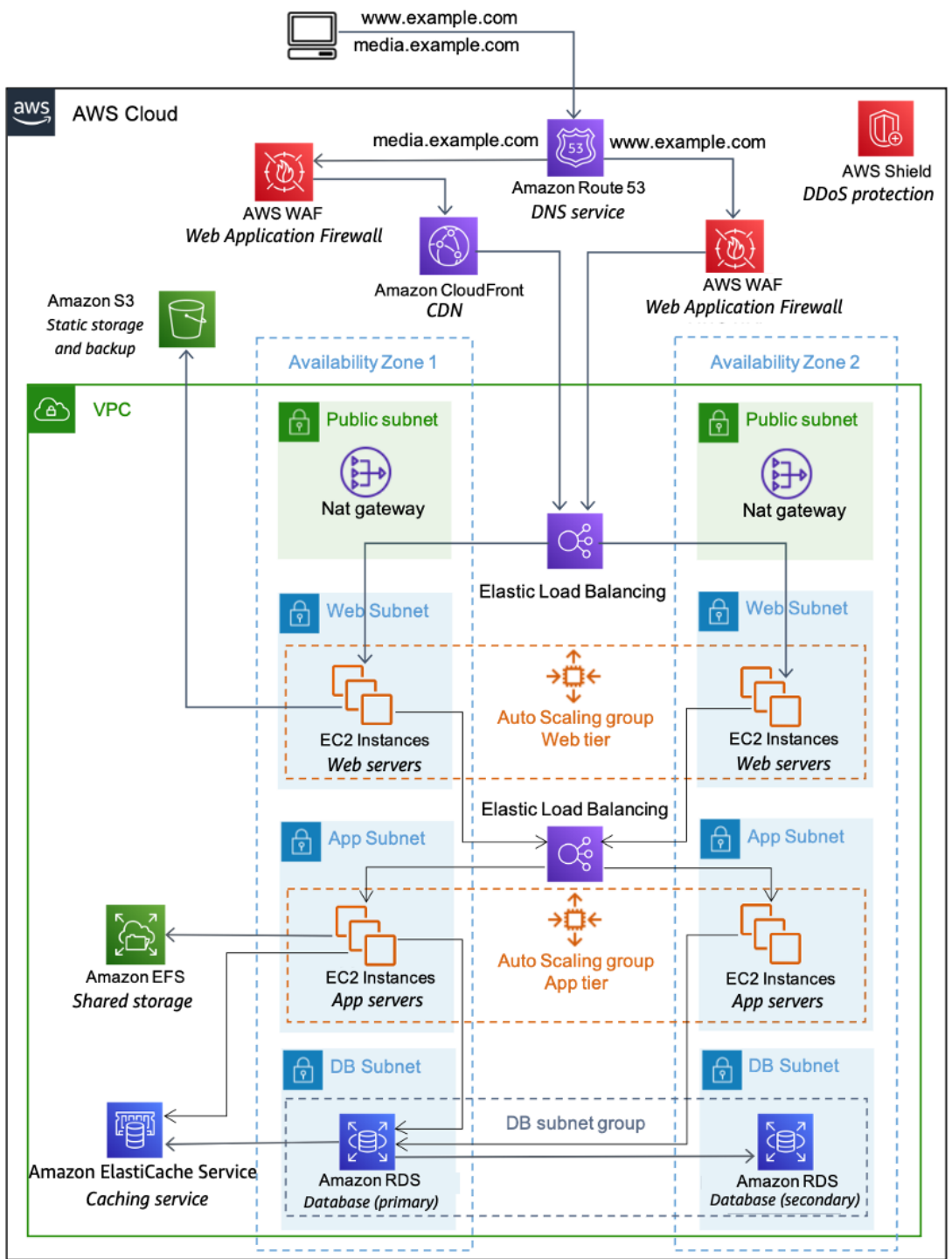
1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)
   - **Software applications** provided as service to the clients
   - Reduce installation/maintenance complexity

Cloud Service Model



Cloud Services (Ex: AWS)

- Computation → EC2 Instance
- Virtual Images → AMI
- Virtual Network → VPC
- Load Balancing → Elastic
- SQL Database → RDS
- NoSQL Database → DynamoDB
- Cache → ElastiCache
- File Storage → EFS
- Object Storage → S3
- Scaling → Auto Scaling



Cloud Services

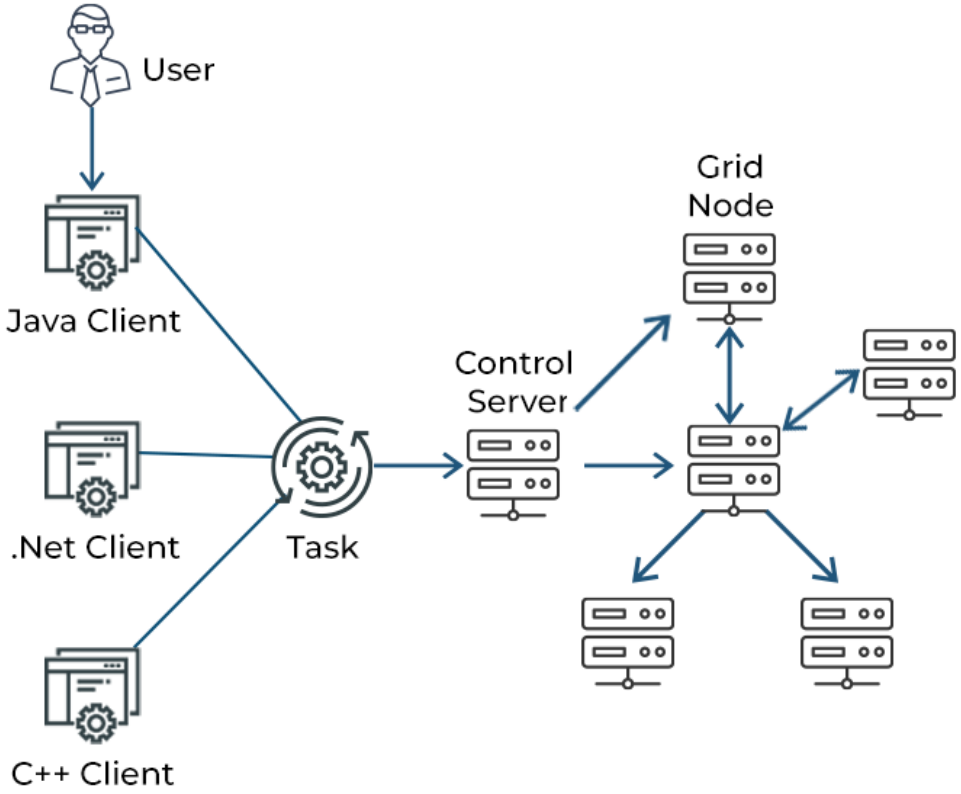
Cloud Services	aws		
Virtual Servers	Instances	VM Instances	VMs
Platform-as-a-Service	Elastic Beanstalk	App Engine	Cloud Services
Serverless Computing	Lambda	Cloud Functions	Azure Functions
Docker Management	ECS	Container Engine	Container Service
Kubernetes Management	EKS	Kubernetes Engine	Kubernetes Service
Object Storage	S3	Cloud Storage	Block Blob
Archive Storage	Glacier	Coldline	Archive Storage
File Storage	EFS	ZFS / Avere	Azure Files
Global Content Delivery	CloudFront	Cloud CDN	Delivery Network
Canaged Data Warehouse	Redshift	Big Query	SQL Warehouse

https://bytebytego.com/guides/cloud-comparison-cheat-sheet/

Related Concepts

Cluster/Grid Computing

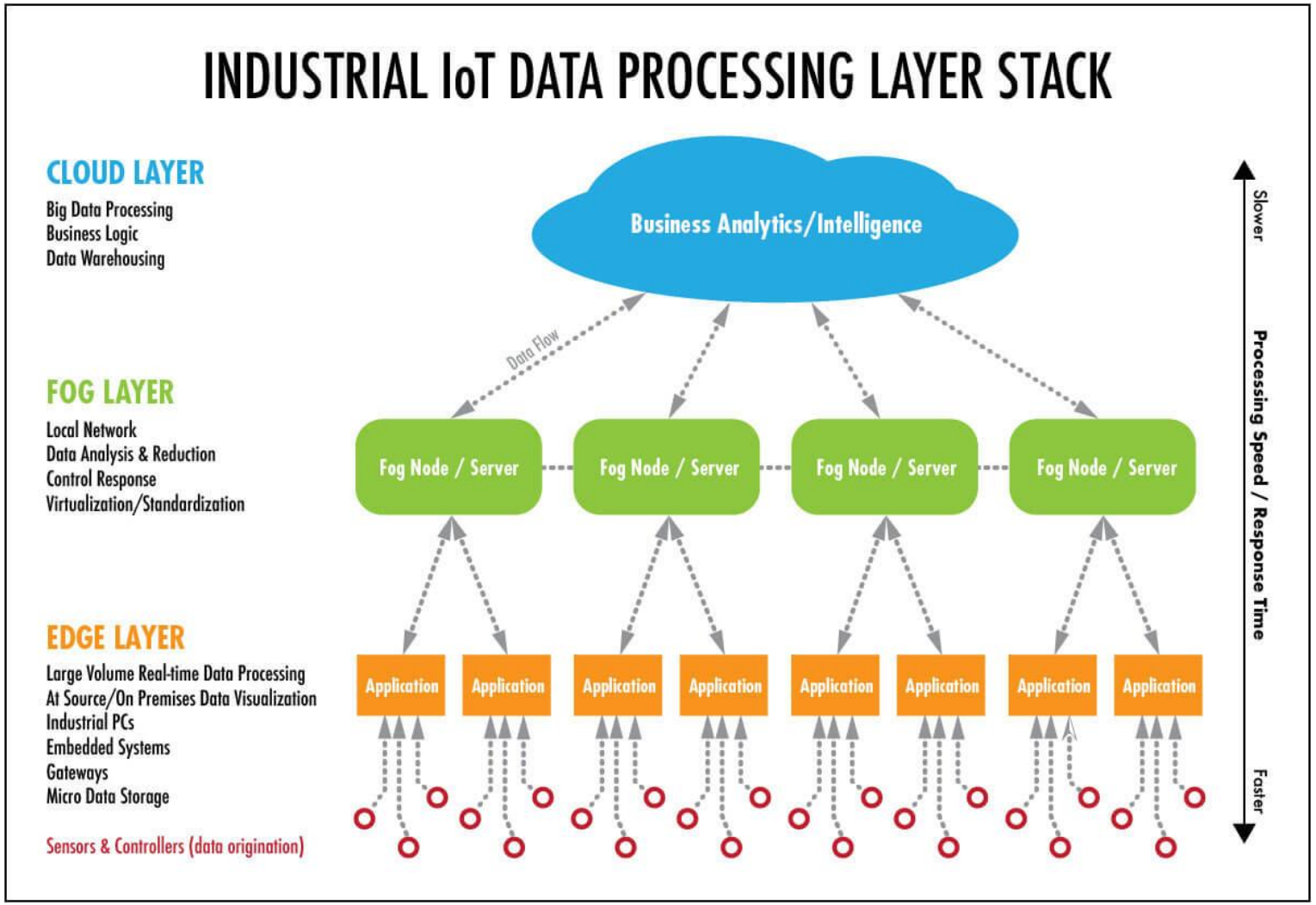
Decentralized heterogeneous (Cluster) / homogeneous (Grid)
Multiple nodes collaborating for a service/application ➔ Bitcoin



IoT/Edge/Fog Computing

Data processing is divided over **multiple layers** for

- Process near source
- Faster response
- Optimize resource
- Cleaner aggregate data



Big Data

Processing stream of data, which is massive in size and speed

Volume → Massive amount of data

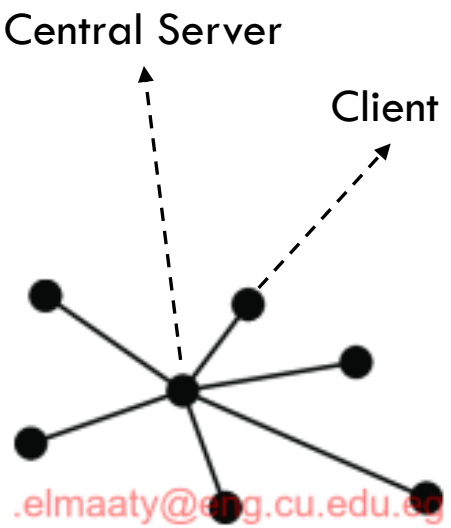
Velocity → Speed of data generation

Variety → Different Types

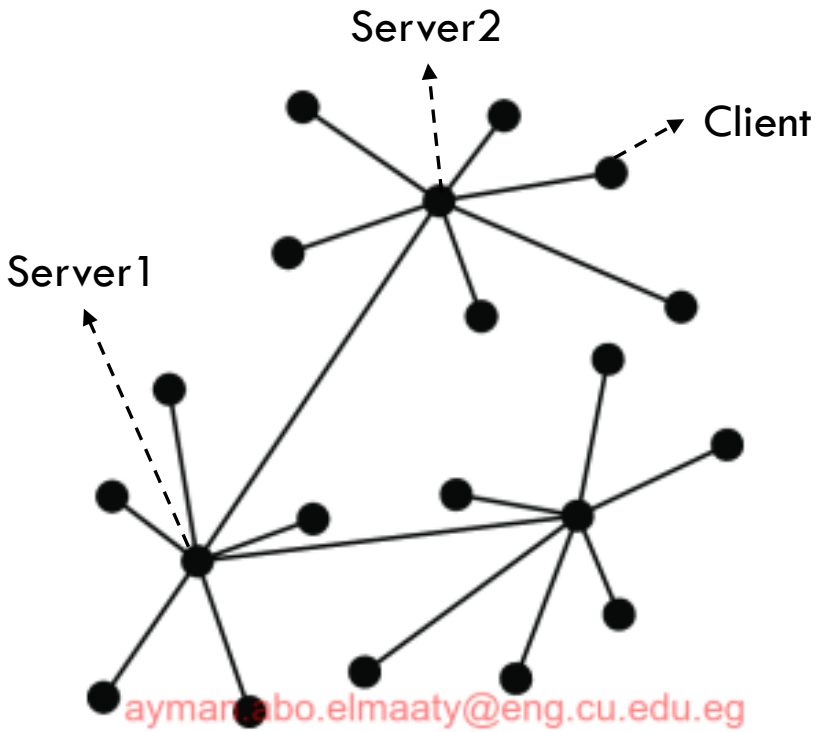
Value → Different usefulness

Veracity → Different Trustworthiness

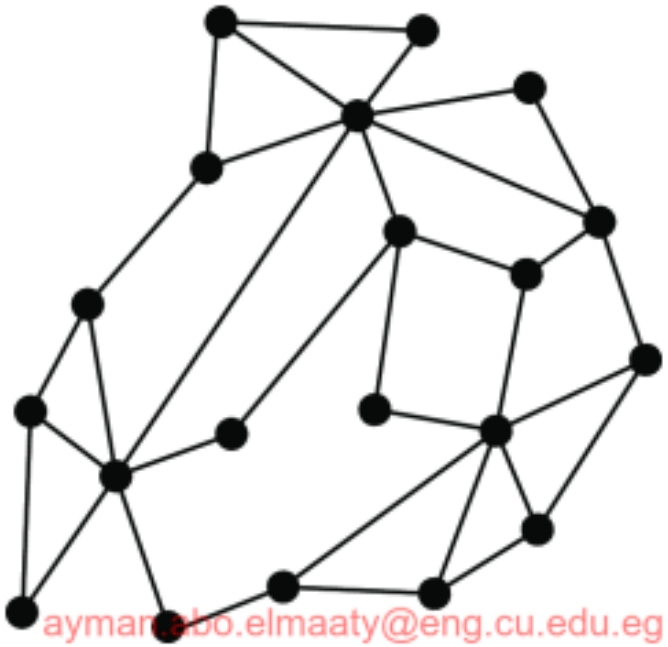
Centralized vs Decentralized vs Distributed



.elmaaty@eng.cu.edu.eg



ayman.abo.elmaaty@eng.cu.edu.eg



ayman.abo.elmaaty@eng.cu.edu.eg

Centralized

1-Server / M-Clients
1-to-1 communication

Decentralized

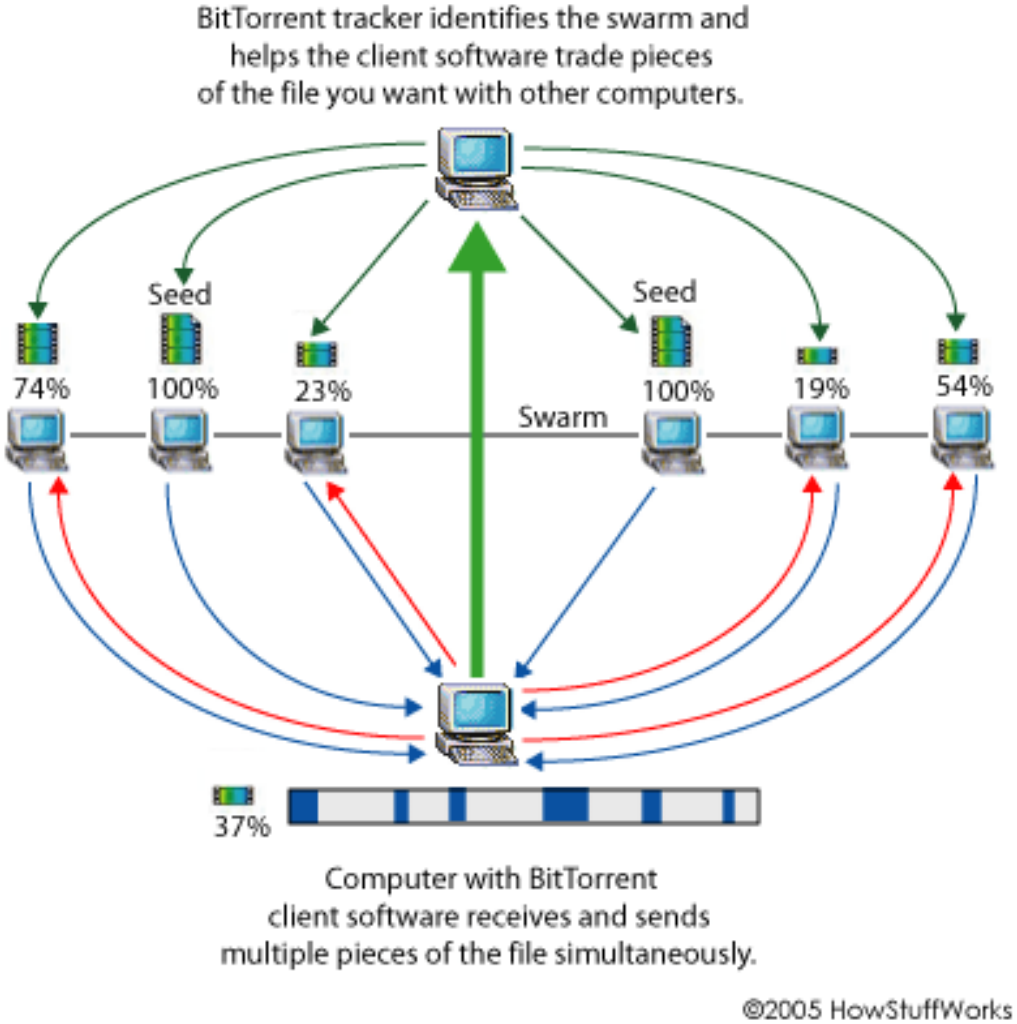
N-Servers / M-Clients
1-to-1 communication

Distributed

N-Servers / M-Clients
1-to-n communication

Peer-to-Peer (P2P)

- Multiple independent components
- Collaborating to perform a service



Cloud Vulnerabilities

Natural Disasters

June 2012

- Storm on East Coast destroyed some Virginia-based Amazon facilities causing prolonged service downtime
- Rebooting process was faulty
  - Restart EC2 (Computation) and EBS (Storage) too extended time
  - Bugs were discovered in Elastic (Load Balancer) and RDS (Database)

Single Point of Failure

November 2025

- Cloudflare outage for ~3 Hrs ➔ 20% Internet down ➔ $60 Billion loss
- Single Config file exceeded limit ➔ Bot mitigation failed
  - Tightly coupled system ➔ BGP Routing protocol failure
- Same buggy code at all ~330 data centers ➔ all data centers failed

Cascading Failures

October 2025

- DynamoDB outage for ~15 hours
  - To be accurate, internal **DNS** had a **race condition**
    - Clean-up deleted old DNS records, that were just applied by a slow DNS enactor

1. DynamoDB can not be reached ➔ Internal services affected
2. EC2 metadata can not be stored ➔ new EC2 instances can not launch
3. Network load balancer can not pass health check ➔ Reduced network capacity
4. Lambda services started failing
5. IAM can not access user data ➔ AWS console unavailable for some users

- Backlog of EC2 requests creating a traffic jam ➔ Slow to reboot

Resources

- Reference Book: Chapter 1
- https://bytebytego.com/guides/cloud-comparison-cheat-sheet/
- https://blog.cloudflare.com/18-november-2025-outage/
- https://medium.com/@spraneeth4/aws-outage-explained-how-dns-and-dynamodb-triggered-a-chain-reaction-197833b8acb1

Thank you